# Q77 AT $N = 2n$: THE $q = 0$ REVERSE SCHUR–WEYL CENTRALISER ALSO FAILS (THE LOCKSTEP LEMMA)

RICK

From: Rick. To: Clio Vega (cc Robin Langer). Date: 2026-11-14.
This note corresponds to WIP commit `69fd79c` (`github.com/grandpa-rick/work-in-progress`).
Follow-up to this morning's note `notes/2026-11-14-q77-reverse-schur-weyl-q0.pdf`.
Proof record (local): `proofs/2026-11-14-q77-N2n-lockstep.md` (PROVE 296); scripts `proofs/prove296/*.py`;
registry `registry/q77-zero-schur-weyl-dc.json` (this repository).

**Grades.**
- proved, cold-read PASS (Dream 295): Theorem 1 ($n \geq 3$, $N = 2n$: $A \subsetneq B$, $\dim B/A \geq 2n-4$). One internal cold read: 0 must-fix, 1 nit (the $k = n$ line in Step 2(ii), included below). Not peer-reviewed.
- computed (exact): defect $\dim B - \dim A = 515 - 513 = 2$ at $(3, 6)$, so the bound is sharp there; the Step 2 claim for all $y \neq y'$, $n = 3, 4, 5$.
- computed (exact $\mathbb{Z}$ certificates): $A \subsetneq B$ at $(4, 7)$ and $(5, 8)$. At $(5, 9)$, $(6, 9)$: the "$\notin A$" half exact, the "$\in B$" half randomised mod $p$.
- hunch: H1, $A = B \iff N \leq n + 2$. open: the B-half support lemma for the conjectured family $a_{n,m}$.

## 1. Setup

$H = H_N(0)$ acts on the right on $V^{\otimes N}$, $\dim V = n$, by $v_w \cdot \pi_i = v_{ws_i}$ if $w_i < w_{i+1}$ (an *ascent*, $i \in \mathrm{Asc}(w)$; we call this a *move*) and $v_w \cdot \pi_i = v_w$ otherwise. $A$ is the image of $H$, $S = \mathrm{End}_H(V^{\otimes N})$, $B = \mathrm{End}_S(V^{\otimes N})$; $A \subseteq B$. We write $\bar{\pi}_i = 1 - \pi_i$, and $\bar{\pi}_{w_0[a,b]}$ for the product of $\bar{\pi}_i$ along a reduced word of the longest element of $\langle s_a, \ldots, s_b \rangle$ (acting on positions $a, \ldots, b+1$). $M_\mu$ is the weight space of weight $\mu$, and $1_\mu \in S$ the weight idempotent.

Let $N = 2n$ and $a_n := \bar{\pi}_{w_0[1,n-1]} \cdot \pi_n \cdot \bar{\pi}_{w_0[n+1,2n-1]}$. For $y \in [n]$ put

$$w_y := 12 \cdots n \cdot y \cdot 23 \cdots n, \qquad \mu_y := \mathrm{wt}(w_y) = (2, \ldots, 2) + e_y - e_1, \qquad u_y := 23 \cdots n \cdot y \cdot 1 \cdot 23 \cdots n.$$

## 2. Theorem

**Theorem 1** (proved, cold-read PASS Dream 295)**.** *Let $n \geq 3$ and $N = 2n$.*

*(1) $a_n \cdot 1_{\mu_y} \in B \setminus A$ for every $y \in \{2, \ldots, n\}$.*
*(2) $\dim B - \dim A \geq 2n - 4$.*

Combined with the theorem of this morning's note ($N \geq 2n + 1$, all $n \geq 2$): $A \subsetneq B$ **for all $n \geq 3$, $N \geq 2n$, and for $n = 2$, $N \geq 5$.** At $(3, 6)$ the bound $2n - 4 = 2$ equals the computed defect $515 - 513 = 2$.

## 3. Proof

B-half. $a_n \cdot 1_C \in B$ for every set $C$ of weights. This is the $N = 2n$ Proposition of the PROVE 295 record (proved; cold-read PASS, Dream 294), already stated in this morning's note.

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*Date*: 2026-11-14.

Step 0 (cyclic-module reduction). Let $s, g$ be words of weights $\mu \neq \mu'$, and suppose $x \in H$ acts on $V^{\otimes N}$ as $\beta$ with $v_g \cdot \beta = 0$. Then $(v_s \cdot \beta, 0) = (v_s, v_g) \cdot x$ lies in

$$X := (v_s, v_g) \cdot H \subseteq M_\mu \oplus M_{\mu'}.$$

$H$ is spanned by the $\pi_x$ ($x \in S_N$), each a product of $\pi_i$'s, and each $\pi_i$ sends a word to a word. So $X$ is spanned by the vectors $e_{(0,a)} + e_{(1,b)}$, $(a, b)$ ranging over the orbit $O$ of $(s, g)$ under *simultaneous* moves (the same $\pi_i$ applied to both). A functional $\lambda$ kills $X$ iff $\lambda(0, a)+\lambda(1, b) = 0$ for all $(a, b) \in O$. (Conversely $X^\perp$ is spanned by signed indicators of connected components of the bipartite graph $O$; only the easy direction is used.)

Step 1 (the coefficient). In $H_m(0)$, $\bar\pi_{w_0} = \sum_{x \in S_m} (-1)^{\ell(x)} \pi_x$.

- Write $\bar\pi_{w_0} = \sum c_x \pi_x$. Expanding the product of $\bar\pi_i = 1 - \pi_i$ along a reduced word gives $c_1 = 1$.
- Every $s_i$ is a right descent of $w_0$, so $\bar\pi_{w_0} = \bar\pi_{w_0 s_i} \bar\pi_i$, and $\bar\pi_i \pi_i = 0$; hence $\bar\pi_{w_0} \pi_i = 0$.
- Expanding, $(\sum c_x \pi_x)\pi_i = \sum_{x < xs_i} (c_x + c_{xs_i})\pi_{xs_i}$, so $c_{xs_i} = -c_x$ for all $x, i$, giving $c_x = (-1)^{\ell(x)}$.
- On a block of $m$ distinct increasing letters, $v_{12\cdots m} \cdot \pi_x = v_x$ (one-line notation), by induction on a reduced word: $\ell(xs_i) > \ell(x) \iff x(i) < x(i+1) \iff$ ascent at $i$. So $v_{12\cdots m} \cdot \bar\pi_{w_0} = \sum_x (-1)^{\ell(x)} v_x$, distinct words, each arrangement $x$ with coefficient exactly $(-1)^{\ell(x)}$.

Now take $y \geq 2$ and $z_y := v_{w_y} \cdot a_n$.

- $\bar\pi_{w_0[1,n-1]}$ acts on positions $1..n$, which hold $12\cdots n$: it gives $\sum_\sigma (-1)^{\ell(\sigma)} v_{\sigma \cdot y \cdot 2\cdots n}$.
- $\pi_n$ swaps positions $n, n+1$ iff $\sigma_n < y$.
- Every later factor acts on positions $\geq n+1$. So a term can contribute to $u_y$ (whose positions $1..n$ read $2\cdots n\, y$) only if position $n$ holds $y$ after $\pi_n$. This forces the swap with $\sigma = 23\cdots n1$, of length $n-1$; the result is $v_{2\cdots n\, y\, 1\, 2\cdots n}$.
- $\bar\pi_{w_0[n+1,2n-1]}$ acts on positions $n+1..2n$, which hold $12\cdots n$; the identity arrangement has coefficient $+1$.

Hence $\langle z_y, v_{u_y} \rangle = (-1)^{n-1} \neq 0$, for every $y \geq 2$.

Step 2 (the lockstep lemma).

**Lemma 2.** *Let $n \geq 3$, $y \neq y'$ in $\{2, \ldots, n\}$, and $O$ the simultaneous-move orbit of $(w_y, w_{y'})$. The only pair in $O$ with first coordinate $u_y$ is $(u_y, u_{y'})$; by symmetry the only pair with second coordinate $u_{y'}$ is $(u_y, u_{y'})$.*

*Proof.* (i) *The path $w_y \to u_y$ is forced.* Label equal letters as instances, left to right; a move swaps two *distinct* letters, so the left-to-right order of the instances of each letter never changes.

- A move adds exactly one (larger-before-smaller) instance pair to the inversion set and changes no other relative order. Inversion sets only grow.
- $\mathrm{inv}(u_y) = \mathrm{inv}(w_y) \sqcup \{(c, 1) : c \text{ an instance at positions } 2..n+1 \text{ of } w_y\}$: the unique 1 has passed $2, \ldots, n$ and $y$, and nothing else changed.
- So along any move path $w_y \to u_y$, every effective move swaps the 1 with its right neighbour. The 1 starts at position 1 and ends at position $n+1$, so the effective moves are, in order, at positions $1, 2, \ldots, n$, with intermediate words

$$a^{(k)} = 2\cdots(k+1)\cdot 1\cdot(k+2)\cdots n\cdot y\cdot 2\cdots n \quad (0 \leq k \leq n-1), \qquad a^{(n)} = u_y.$$

(ii) *Equal ascent sets.* Let $b^{(k)}$ be $a^{(k)}$ with the instance $y$ replaced by $y'$. They differ in one position $p$ ($p = n+1$ for $k \leq n-1$, $p = n$ for $k = n$), so the ascent status can differ only at $p-1$ and $p$.

- $k \leq n-2$: the neighbours of $p$ are $n$ (left) and 2 (right). Since $2 \leq y, y' \leq n$, both positions are non-ascents in both words.
- $k = n-1$: the neighbours are 1 (left; $1 < y, y'$, an ascent in both) and 2 (right; a non-ascent in both).

- $k = n$ ($p = n$): the neighbours are $n$ (left, at position $n-1 \geq 1$; a non-ascent in both) and 1 (right; a non-ascent in both). [This case is needed in (iii) for stutters after the first coordinate has reached $u_y$.]

So $\mathrm{Asc}(a^{(k)}) = \mathrm{Asc}(b^{(k)})$ for $0 \leq k \leq n$. This uses $y' \geq 2$: for $y' = 1$ position $p$ would be an ascent in $b$ but not in $a$.

(iii) *Induction along a common move sequence.* Take any sequence of simultaneous $\pi_i$'s from $(w_y, w_{y'}) = (a^{(0)}, b^{(0)})$ ending with first coordinate $u_y$. Its first coordinates form a move path $w_y \to u_y$ with stutters; by (i) each actual change is the next 1-move. Suppose the current pair is $(a^{(k)}, b^{(k)})$ and $\pi_i$ is applied. If $i \notin \mathrm{Asc}(a^{(k)}) = \mathrm{Asc}(b^{(k)})$, both words are fixed. If $i \in \mathrm{Asc}$, both swap positions $i, i+1$; by (i) $i = k+1$, and since the words differ only by $y \leftrightarrow y'$ at a common position, the new pair is $(a^{(k+1)}, b^{(k+1)})$. At the end the pair is $(a^{(n)}, b^{(n)}) = (u_y, u_{y'})$.

The second claim is the same argument for the flipped orbit of $(w_{y'}, w_y)$. □

Step 3 ($a_n \cdot 1_{\mu_y} \notin A$). Fix $y \in \{2, \ldots, n\}$ and pick $y' \in \{2, \ldots, n\} \setminus \{y\}$ (possible as $n \geq 3$). Suppose $a_n \cdot 1_{\mu_y}$ is the action of some $x \in H$. Since $\mu_{y'} \neq \mu_y$, $v_{w_{y'}} \cdot x = 0$, so $(z_y, 0) \in X = (v_{w_y}, v_{w_{y'}}) \cdot H$. Let $\lambda := \delta_{(0,u_y)} - \delta_{(1,u_{y'})}$. By Lemma 2, $\lambda$ vanishes on every spanning vector $e_{(0,a)} + e_{(1,b)}$, $(a,b) \in O$: either $(a,b) = (u_y, u_{y'})$, giving $1 - 1 = 0$, or neither coordinate is hit. So $\lambda(X) = 0$, but $\lambda(z_y, 0) = (-1)^{n-1} \neq 0$ by Step 1. Contradiction. This proves (1).

Step 4 (the lower bound).

- *Mirror.* Let $\theta(v_w) := v_{w^*}$, $w_i^* := n + 1 - w_{N+1-i}$. Then $\theta$ maps ascents at $i$ to ascents at $N - i$ and moves to moves, so $\theta(v \cdot \pi_i) = \theta(v) \cdot \pi_{N-i}$. As $\pi_i \mapsto \pi_{N-i}$ is an automorphism of $H$, $\theta A \theta^{-1} = A$, $\theta S \theta^{-1} = S$, $\theta B \theta^{-1} = B$. Put $\beta_y := a_n \cdot 1_{\mu_y}$ and $\beta_y^* := \theta \beta_y \theta^{-1}$ ($y = 2..n$). Then $\beta_y^* \in B \setminus A$, supported on the reversed weight $\mu_y^{\mathrm{rev}}$.
- *Distinct weights.* $\mu_y^{\mathrm{rev}}$ has the letter $n$ exactly once, while $\mu_y$ has $n$ at least twice; so the $2(n-1)$ weights $\mu_y, \mu_y^{\mathrm{rev}}$ are pairwise distinct.
- *Independence.* Each $\beta$ is nonzero (Step 1), so the $2(n-1)$ elements are linearly independent.
- *Constancy.* Suppose $x := \sum c_y \beta_y + \sum d_y \beta_y^* \in A$. For $y \neq y'$ in $[2, n]$, $x$ acts on $M_{\mu_y}$ as $c_y \beta_y$ and on $M_{\mu_{y'}}$ as $c_{y'} \beta_{y'}$, so $(c_y z_y, c_{y'} z_{y'}) \in X$, and $\lambda$ gives $(-1)^{n-1}(c_y - c_{y'}) = 0$. So $c$ is constant; applying $\theta$, $d$ is constant too.

Hence $\mathrm{span}\{\beta_y, \beta_y^*\} \cap A$ has dimension $\leq 2$, and $\dim B/A \geq 2(n-1) - 2 = 2n - 4$. This proves (2). □

**Remark 3** (Why $n = 2$ is different)**.** For $n = 2$ the only $y \geq 2$ is $y = 2$, so there is no partner $y'$; and $y' = 1$ fails Step 2(ii) ($w_1$ has an ascent at $n+1$, $w_2$ does not). Consistent with $A = B$ at $(2, 4)$.

**Remark 4** (A reusable certificate)**.** Step 0 certifies "$\beta \notin A$" for a single-weight witness by *one* partner word $g$ plus the connected components of the bipartite simultaneous-move orbit graph of $(s, g)$; no rank computation. Picture: on the cyclic module generated by the two words (common ascent set $[1, n-1] \cup [n+2, 2n-1]$) the vectors $u_y, u_{y'}$ are glued, so $A$ reaches them only together; $1_{\mu_y} \in S$ separates them, and $B$ can too once $a_n$ has killed every cross-weight image.

## 4. Computed evidence (exact; `proofs/prove296/`)

- `single.py`: exhaustive single-partner search. At $(3, 6)$, $y = 2$, exactly one word certifies $(z, 0) \notin (v_{w_y}, v_g)H$, namely $g = w_3 = 123323$; $y = 3 \mapsto g = w_2$. At $(4, 8)$ (same-Asc candidates), $y = 2 \mapsto \{w_3, w_4\}$, $y = 3 \mapsto \{w_2, w_4\}$. The lockstep partners are exactly the certificates.
- `comp.py`: at $(3, 6)$ every one of the 12 words of $z$ is in its own bipartite component, with $z$-sum $\pm 1$.
- `ucomp.py`: Lemma 2 (both directions) and $\langle z_y, u_y \rangle = (-1)^{n-1}$ for all ordered $y \neq y' \in [2, n]$, $n = 3, 4, 5$: 20 cases, orbits up to 73440 pairs.

- Sharpness at $(3,6)$: $\dim A = 513$, $\dim B = 515$, defect $2 = 2n-4$.

## 5. Beyond the theorem: the window $n+3 \le N \le 2n-1$

- computed, exact: $(4,7)$ fails. Witness $a = \bar\pi_{w_0[1,3]}\,\pi_4\,\bar\pi_{w_0[5,6]}\,\bar\pi_3\bar\pi_4\bar\pi_5$ on $\mu = (1,1,2,3)$. $B$: every cross-image word is killed, over $\mathbb{Z}$. $\notin A$: Step 0 with partner 1234334. So the hunch "$A = B \iff N \le 2n-1$" from this morning's note is **dead**.
- computed, exact: $(5,8)$ fails. At $(5,9)$ and $(6,9)$ the same family gives a witness with the "$\notin A$" half exact and the "$\in B$" half randomised mod $p$.
- hunch H1: $A = B \iff N \le n+2$. Supporting data: $A = B$ computed at $(3,5)$, $(4,6)$ and smaller cases.
- conjecture (computed on all tested cases): for $N = n+m$, $3 \le m \le n$, $a_{n,m} = \bar\pi_{w_0[1,n-1]}\,\pi_n\,\bar\pi_{w_0[n+1,n+m-1]}\,\bar\pi_x$, where $x$ has the reduced word of $m$ descending runs of length $n-m$; source $12\cdots n\cdot n\cdot(n-m+2\cdots n)$, partner $12\cdots n\cdot(n-1)\cdot(n-m+2\cdots n)$. $m=n$ is Theorem 1. It would give $A \subsetneq B$ for all $N \ge n+3$.
- open: the lockstep half generalises (it needs exactly $m \ge 3$), but the B-half needs a support lemma for the tail $\bar\pi_x$, which is **open**. (Computed support: $\{12\cdots n\cdot T : T \text{ an increasing } m\text{-subset}\} \cup \{12\cdots n\cdot y\cdot H\}$, $H = (n-m+2,\dots,n)$, $y \in H$.)
- open: the equality half of H1 ($N = n+1, n+2$); no proof idea yet.